



Student Name: _____

Saint Ignatius' College
Riverview

2003
YEAR 12
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Time allowed – 2 hours,
+ 5 minutes reading time
- Write using blue or black pen
- Board-approved calculators and
mathaids may be used
- Show all necessary working
- Answer each question in a
separate booklet with your name
and teacher's name

Total Marks (84)

- Attempt Questions 1 – 7

Total marks (84)

Attempt Questions 1 – 7

All questions are of equal value

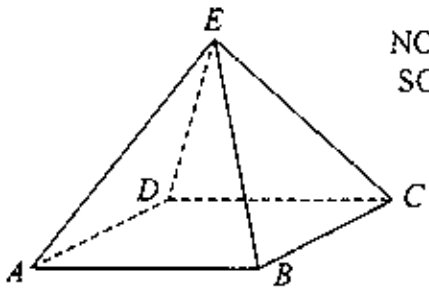
Answer each question in a SEPARATE writing booklet.

QUESTION 1 (12 Marks) Use a SEPARATE writing booklet. **Marks**

(a) Find the acute angle between the lines $y = 2x - 5$ and $y = 6 - 3x$. **2**

(b) Solve $\frac{x+4}{x} < 3$. **3**

(c) Find the general solutions of the equation $\sin 2\theta = \sin^2 \theta$. **4**
Give your answer in terms of π .

(d)  **NOT TO SCALE** **3**

A right square pyramid $ABCDE$ has a base of length 6cm and a perpendicular height of 8cm.

Find the angle which the slant edge AE makes with the base $ABCD$.

QUESTION 2 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) The point $(2, 2)$ divides the join of $(-2, 5)$ to (a, b) in the ratio $3:2$.
Find the values of a and b .

2

- (b) If α, β, γ are the roots of the equation $2x^3 - 6x^2 + 5x - 1 = 0$,
find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

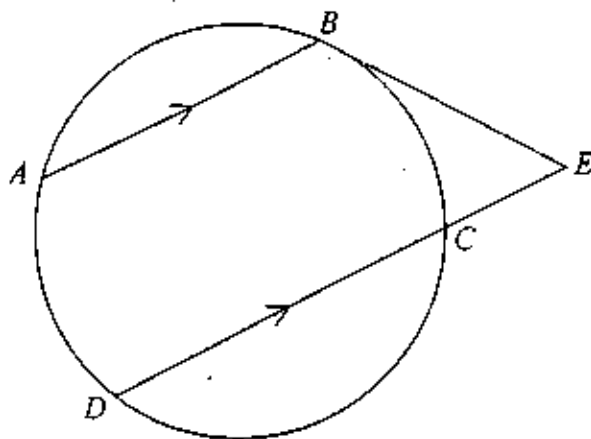
3

- (c) The polynomial equation $x^3 - 11x^2 + px + q = 0$ has a double root
at $x = \alpha$ and a single root at $x = \alpha + 2$.
Using the formula for the sum of the roots, or otherwise, find the
values of α , p and q .

4

- (d)

3



In the diagram, A, B, C and D lie on a circle.

AB is parallel to DC and the tangent at B meets DC produced at E .

Copy or trace the diagram onto your writing page, and join BC and AC .

Prove that $\triangle ABC$ is similar to $\triangle BCE$.

QUESTION 3 (12 Marks) Use a SEPARATE writing booklet. **Marks**

- (a) Find the inverse function of the function $f(x) = \frac{5-2x}{3}$, expressing your answer in the form $f^{-1}(x) = \dots$. **2**
- (b) Evaluate $\cos^{-1}\left(\frac{1}{2} \tan \frac{2\pi}{3}\right)$. **2**
- (c) Find the exact value of $\sin\left(2 \cos^{-1} \frac{2}{3}\right)$. **3**
- (d) Prove $\frac{\cos A - \sin A}{\cos A + \sin A} = \frac{\cos 2A}{1 + \sin 2A}$. **2**
- (e) Show that there is only one stationary point on the curve $y = x + \cos^{-1} x$, and determine its nature. **3**

QUESTION 4 (12 Marks) Use a SEPARATE writing booklet.

Marks

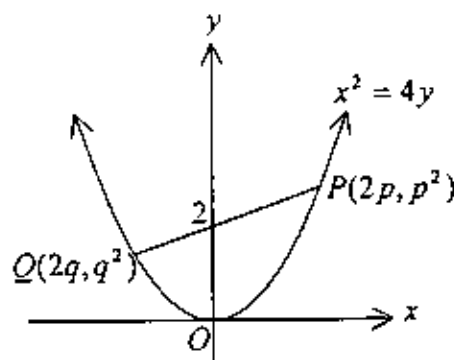
(a) From a standard pack of 52 cards, a hand of 4 cards is dealt.

- | | | |
|------|---|---|
| (i) | How many different hands can be selected? | 1 |
| (ii) | What is the probability I will be dealt exactly two aces? | 2 |

(b) The letters of the word CALCULUS are arranged in a row.

- | | | |
|------|--|---|
| (i) | How many different arrangements are possible? | 2 |
| (ii) | In how many of the arrangements will the letters U be at each end? | 1 |

(c)



Points $P(2p, p^2)$ and $Q(2q, q^2)$ lie on the parabola $x^2 = 4y$.

- | | | |
|-------|--|---|
| (i) | Show that the equation of the chord PQ is $(p + q)x - 2y - 2pq = 0$. | 2 |
| (ii) | Find the coordinates of M , the midpoint of PQ . | 1 |
| (iii) | Hence find the equation of the locus of M if the chord PQ crosses the y axis at $(0, 2)$. | 3 |

QUESTION 5 (12 Marks) Use a SEPARATE writing booklet.

Marks

(a) Find the following indefinite integrals:

(i) $\int \frac{1}{4+x^2} dx$ 1

(ii) $\int \frac{x}{4+x^2} dx$ 1

(b) Find $\int_0^{\pi} \sin^2 x dx$. 3

(c) Find $\int \frac{e^{2x}}{e^x - 2} dx$ using the substitution $u = e^x - 2$. 3

(d) The acceleration of a particle moving in a straight line at position x is given by $\ddot{x} = -\frac{6}{(x+1)^2}$. Initially it has velocity 4 units when it is at the origin. 4

Show that the velocity v at position x is given by $v = \pm 2\sqrt{\frac{x+4}{x+1}}$.

QUESTION 6 (12 Marks) Use a SEPARATE writing booklet. **Marks**

(a) Use the table of standard integrals to find $\int \frac{1}{\sqrt{x^2 + 16}} dx$. 1

(b) Prove by mathematical induction, for positive integers n , that 4

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

(c) Use one application of Newton's method to find an approximation to the root of $2x - 4 \sin 3x = 0$ near $x = 1$. 3
Write your answer to two decimal places.

(d) (i) On the same set of axes, sketch the graphs of the equations 2
 $y = |2x|$ and $y = x^2 - 3$.

(ii) Hence or otherwise solve the inequality 2
 $|2x| > x^2 - 3$.

QUESTION 7 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) An object is projected from level ground at an angle θ to the horizontal, with a velocity of V m/s. The object returns to the ground after 4 seconds and 100 metres from its point of projection. Assume acceleration due to gravity is 10 m/s^2 , and neglect air resistance.
- (i) From the equations for acceleration in the x and y directions, find expressions for x and y in terms of time t ($t \leq 4$). 2
- (ii) Hence find the values of V and θ . 2
- (iii) What is the maximum height reached by the object? 2

- (b) Newton's Law of Cooling states that the rate at which a body cools is proportional to the difference between the temperature of the body and that of the surrounding medium.

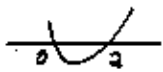
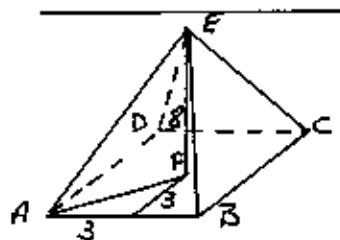
$$\text{i.e. } \frac{dT}{dt} = -k(T - T_0)$$

where T is the temperature of the body at time t and T_0 is the temperature of the surrounding medium, assumed constant.

- (i) Show that $T = T_0 + Ae^{-kt}$ is a solution to this equation. 1
- (ii) A body whose temperature is 150°C is cooled by placing it in a liquid at 25°C . In one minute, the temperature of the body had cooled to 100°C . 5
How long will it take for the body to cool to 50°C ?

End of paper

Mathematics Extension 1: Question

Suggested Solutions	Marks Awarded	Marker's Comments
(a) $y = 2x - 5$ $y = 6 - 3x$ $\tan \theta = \left \frac{2 - (-3)}{1 + 2 \times (-3)} \right \quad \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)$ $= 1$ $\theta = 45^\circ$ (2) (b) $\frac{x+4}{x} < 3$ $x(x+4) < 3x^2$ $2x^2 - 4x > 0$ $2x(x-2) > 0$  $x < 0, x > 2$ (3) (c) $\sin 2\theta = \sin^2 \theta$ $2 \sin \theta \cos \theta - \sin^2 \theta = 0$ $\sin \theta (2 \cos \theta - \sin \theta) = 0$ $\sin \theta = 0$ or $\sin \theta = 2 \cos \theta$ $\tan \theta = 2$ $\theta = n\pi, \theta = n\pi + \tan^{-1} 2$ OR $n\pi + 1.11$ (2dp) (4) (d)  $AF = \sqrt{3^2 + 3^2} = \sqrt{18}$ $\tan \theta = \frac{EF}{AF} = \frac{8}{\sqrt{18}}$ $\theta = 62^\circ 04'$ OR 62° (nearest degree) (3)		

Mathematics Extension 1: Question 2

Suggested Solutions

Marks
Awarded

Marker's Comments

2(a) $\begin{matrix} x_1 & y_1 & x_2 & y_2 & m_1 & m_2 \\ (-2, 5) & (a, b) & 3 & 2 \end{matrix}$

$$\frac{3a + 2(-2)}{3+2} = 2; \quad \frac{3 \times b + 2 \times 5}{3+2} = 2$$

$$a = 4\frac{2}{3}, \quad b = 0 \quad (2)$$

(b) $2x^3 - 6x^2 + 5x - 1 = 0$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \beta\alpha}{\alpha\beta\gamma}$$

$$= \frac{\frac{5}{2}}{\frac{1}{2}} = 5 \quad (3)$$

(c) $x^3 - 11x^2 + px + q = 0$

Roots are $\alpha, \alpha, \alpha+2$.

Sum of roots: $\alpha + \alpha + (\alpha+2) = 11$

$$\alpha = 3.$$

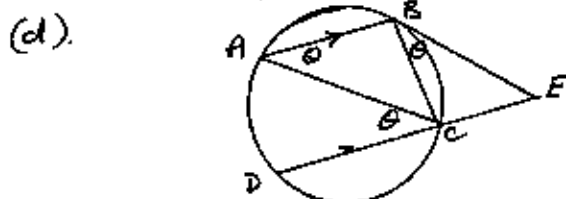
$\therefore$ Roots are 3, 3, 5.

$\alpha\beta + \alpha\gamma + \beta\gamma$: $3 \times 3 + 3 \times 5 + 3 \times 5 = p$

$$p = 39.$$

$\alpha\beta\gamma$: $3 \times 3 \times 5 = -q$

$$q = -45. \quad (4)$$



$\angle CBE = \angle BAC = \theta$ (alt. segment thm) (A)

$\angle BAC = \angle ACD = \theta$ (alt. angles $AB \parallel DC$)

$\angle BCD = \angle CBE + \angle BCE$ (ext. angle of $\triangle BCE$)

$\theta + \angle ACB = \theta + \angle BCE$

$\therefore \angle ACB = \angle BCE$ (B)

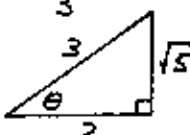
From (A), (B), two pairs of angles are equal

$\therefore \triangle ABC \parallel \triangle BCE \quad (3)$

Alternative:

$\angle ABC = \angle BCE$ (alternate $\angle$ s in parallel lines)

Mathematics Extension 1: Question 3

Suggested Solutions	Marks Awarded	Marker's Comments
3.(a) Let $y = \frac{5-2x}{3}$ Inverse is: $x = \frac{5-2y}{3}$ $3x = 5-2y$ $y = \frac{5-3x}{2}$ $\therefore f^{-1}(x) = \frac{5-3x}{2}$ (2)		
(b) $\cos^{-1}\left(\frac{1}{2} \tan \frac{2\pi}{3}\right) = \cos^{-1}\left(\frac{1}{2} \times -\sqrt{3}\right)$ $= \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ $= \frac{5\pi}{6}$ (2)		
(c) $\sin\left(2 \cos^{-1} \frac{2}{3}\right)$ Let $\theta = \cos^{-1} \frac{2}{3}$ $= \sin 2\theta$ $\cos \theta = \frac{2}{3}$ $= 2 \sin \theta \cos \theta$ $= 2 \times \frac{\sqrt{5}}{3} \times \frac{2}{3}$ $= \frac{4\sqrt{5}}{9}$ (3) 		
(d) $\frac{\cos A - \sin A}{\cos A + \sin A} = \frac{\cos A - \sin A}{\cos A + \sin A} \times \frac{\cos A + \sin A}{\cos A + \sin A}$ $= \frac{\cos^2 A - \sin^2 A}{\cos^2 A + \sin^2 A + 2 \sin A \cos A}$ $= \frac{\cos 2A}{1 + \sin 2A}$ (2)		
(e) $y = x + \cos^{-1} x$ $\frac{dy}{dx} = 1 - \frac{1}{\sqrt{1-x^2}} = 1 - (1-x^2)^{-\frac{1}{2}}$ For stat. point: $1 = \frac{1}{\sqrt{1-x^2}} \therefore \sqrt{1-x^2} = 1$ $\therefore x = 0$ $\frac{d^2y}{dx^2} = \frac{1}{2} (1-x^2)^{-\frac{3}{2}} (-2x) = \frac{-x}{(1-x^2)^{3/2}}$ When $x=0$, $\frac{d^2y}{dx^2} = 0$. If $x < 0$, $\frac{d^2y}{dx^2} > 0$; If $x > 0$, $\frac{d^2y}{dx^2} < 0$ Concavity changes $\therefore$ one stationary point is a horizontal point of inflexion. (3)		

Mathematics Extension 1: Question 4

Suggested Solutions

Marks
Awarded

Marker's Comments

$$4.(a)(i) \text{ No. of different hands} = \binom{52}{4} \\ = 270\,725 \quad (1)$$

$$(ii) P(2 \text{ aces}) = \frac{\binom{4}{2} \binom{48}{2}}{\binom{52}{4}} \\ = 0.025 \quad (2)$$

$$(b)(i) \text{ No. of arrangements} = \frac{8!}{2!2!2!} \\ = 5040 \quad (2)$$

$$(ii) \text{ No. arrgts. with U at ends} = \frac{6!}{2!2!} \\ = 180 \quad (1)$$

$$(c)(i) PQ: \frac{y - q^2}{x - 2q} = \frac{p^2 - q^2}{2p - 2q} = \frac{p + q}{2} \\ 2y - 2q^2 = (p + q)x - 2q(p + q) \\ 2y - 2q^2 = (p + q)x - 2pq - 2q^2 \\ (p + q)x - 2y - 2pq = 0 \quad (2)$$

$$(ii) M: \left(\frac{2p + 2q}{2}, \frac{p^2 + q^2}{2} \right) \\ \text{i.e. } (p + q, \frac{p^2 + q^2}{2}) \quad (1)$$

$$(iii) \text{ If } PQ \text{ passes through } (0, 2) \\ \text{Subst. } x = 0, y = 2: 0 - 2 \times 2 - 2pq = 0 \\ pq = -2$$

$$\therefore x = p + q, \quad y = \frac{p^2 + q^2}{2} \\ (p + q)^2 = p^2 + q^2 + 2pq \\ x^2 = 2y + 2 \times (-2) \\ x^2 = 2y - 4$$

$$\text{Locus of } M \text{ is } x^2 = 2y - 4. \quad (3)$$

Mathematics Extension 1: Question 5

Suggested Solutions	Marks Awarded	Marker's Comments
5. (a)(i) $\int \frac{1}{4+x^2} dx = \frac{1}{2} \tan^{-1} \frac{x}{2} + C$ (1) (ii) $\int \frac{x}{4+x^2} dx = \frac{1}{2} \int \frac{2x}{4+x^2} dx$ $= \frac{1}{2} \log_e(4+x^2) + C$ (1) (b) $\int_0^{\frac{\pi}{6}} \sin^2 x dx = \int_0^{\frac{\pi}{6}} \frac{1}{2} (1 - \cos 2x) dx$ $= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{6}}$ $= \frac{1}{2} \left[\left(\frac{\pi}{6} - \frac{1}{2} \times \frac{\sqrt{3}}{2} \right) - (0 - 0) \right]$ $= \frac{\pi}{12} - \frac{\sqrt{3}}{8} \text{ or } \frac{2\pi - 3\sqrt{3}}{24}$ (3)		
(c) $\int \frac{e^{2x}}{e^x - 2} dx$ $= \int \frac{e^x e^x dx}{e^x - 2}$ $= \int \frac{(u+2) du}{u}$ $= \int \left(1 + \frac{2}{u} \right) du$ $= u + 2 \ln u + C$ $= e^x - 2 + 2 \ln(e^x - 2) + C$ (3) $u = e^x - 2$ $\frac{du}{dx} = e^x$ $du = e^x dx$		
(d) $\ddot{x} = \frac{-6}{(x+1)^2}$ $\frac{d}{dt} \left(\frac{1}{2} v^2 \right) = -6 (x+1)^{-2}$ $\frac{1}{2} v^2 = 6 (x+1)^{-1} + C$ When $x=0, v=4: 8 = 6 + C$ $C = 2$ $\therefore \frac{1}{2} v^2 = \frac{6}{x+1} + 2$ $= \frac{6 + 2(x+1)}{x+1}$ $= \frac{2x+8}{x+1}$ $= 2 \left(\frac{x+4}{x+1} \right)$ $v^2 = 4 \left(\frac{x+4}{x+1} \right)$ $v = \pm 2 \sqrt{\frac{x+4}{x+1}}$ (4)		

Mathematics Extension 1: Question 6

Suggested Solutions

Marks
Awarded

Marker's Comments

$$6(a) \int \frac{1}{\sqrt{x^2+16}} dx = \log_e(x + \sqrt{x^2+16}) + C \quad (1)$$

$$(b) \text{ Prove } \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

$$\text{When } n=1, \text{ LHS} = \frac{1}{1 \cdot 3} = \frac{1}{3}, \text{ RHS} = \frac{1}{2+1} = \frac{1}{3}$$

$\therefore$ it is true for $n=1$.

Assume it is true for $n=k$.

$$\text{i.e. assume } \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1}$$

When $n=k+1$,

$$\text{LHS} = \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)}$$

$$= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \text{ by assumption}$$

$$= \frac{k(2k+3) + 1}{(2k+1)(2k+3)}$$

$$= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)}$$

$$= \frac{(k+1)(2k+1)}{(2k+1)(2k+3)}$$

$$= \frac{k+1}{2(k+1)+1} = \frac{n}{2n+1} \text{ where } n=k+1.$$

$\therefore$ if it is true for $n=k$, it is true for $n=k+1$.

Since it is true for $n=1$, it is true for $n=2, n=3, \dots$

(4)

$$(c) \text{ Let } f(x) = 2x - 4 \sin 3x$$

$$f'(x) = 2 - 12 \cos 3x$$

$$f(1) = 2 - 4 \sin 3 = 1.4355$$

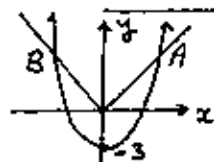
$$f'(1) = 2 - 12 \cos 3 = 13.880$$

$$\text{Approx'n} = 1 - \frac{f(1)}{f'(1)} = 1 - \frac{1.4355}{13.880}$$

$$= 0.90 \text{ (2dp)}$$

(3)

(d)(i)



(2)

$$(ii) x^2 - 3 = 2x$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3, -1$$

$$A(3, 6) \quad B(-3, 6)$$

$$\therefore |2x| > x^2 - 3 \text{ for } -3 < x < 3.$$

(2)

Mathematics Extension 1: Question 7(a)

Suggested Solutions

Marks
Awarded

Marker's Comments

7(a) (i) $\ddot{x} = 0$

$\dot{x} = c$

When $t=0$, $\dot{x} = V \cos \theta$

$\therefore c = V \cos \theta$

$\therefore \dot{x} = V \cos \theta$

$x = V \cos \theta t + c'$

When $t=0$, $x=0 \therefore c'=0$

$\therefore x = V \cos \theta t$

$\ddot{y} = -10$

$\dot{y} = -10t + k$

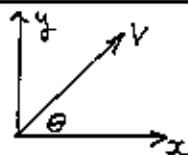
When $t=0$, $\dot{y} = V \sin \theta \therefore k = V \sin \theta$

$\therefore \dot{y} = V \sin \theta - 10t$

$y = V \sin \theta t - 5t^2 + k'$

When $t=0$, $y=0 \therefore k'=0$

$\therefore y = V \sin \theta t - 5t^2$ (2)



(ii) When $t=4$, $y=0$, $x=100$

$100 = 4V \cos \theta \quad 0 = 4V \sin \theta - 80$

$V \cos \theta = 25 \quad V \sin \theta = 20$

$\frac{V \sin \theta}{V \cos \theta} = \frac{20}{25} \therefore \tan \theta = 0.8$

$\theta = 38^\circ 40'$

Also, $V^2 \cos^2 \theta + V^2 \sin^2 \theta = 25^2 + 20^2$

$V^2 (\cos^2 \theta + \sin^2 \theta) = 1025$

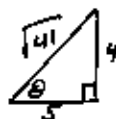
$V = \sqrt{1025} \text{ or } 32.0 \text{ m/s}$ (2)

$= 5\sqrt{41}$

(iii) Maximum height when $\dot{y}=0$

$5\sqrt{41} \sin 38^\circ 40' - 10t = 0$

$t=2$



When $t=2$, $y = 5\sqrt{41} \times \frac{4}{\sqrt{41}} \times 2 - 5 \times 2^2$

$= 20$

Maximum height is 20 m.

(2)

Mathematics Extension 1: Question 7(b)

Suggested Solutions

Marks
Awarded

Marker's Comments

$$7.(b) \quad \frac{dT}{dt} = -k(T - T_0)$$

$$(i) \quad T = T_0 + A e^{-kt}$$

$$\frac{dT}{dt} = -k A e^{-kt}$$

$$= -k(T - T_0)$$

(1)

$$(ii) \text{ When } t=0, T=150, T_0=25$$

$$150 = 25 + A$$

$$\therefore A = 125$$

✓

$$\therefore T = 25 + 125 e^{-kt}$$

$$\text{When } t=1, T=100$$

$$100 = 25 + 125 e^{-k}$$

$$75 = 125 e^{-k}$$

$$e^k = \frac{125}{75}$$

$$k = \ln\left(\frac{125}{75}\right)$$

$$= 0.5108 \text{ (4dp)} \quad \checkmark \checkmark$$

$$\therefore T = 25 + 125 e^{-0.5108t}$$

$$\text{When } T=50,$$

$$50 = 25 + 125 e^{-0.5108t}$$

$$25 = 125 e^{-0.5108t}$$

$$e^{0.5108t} = \frac{125}{25} = 5$$

$$0.5108t = \ln 5$$

$$t = \frac{\ln 5}{0.5108}$$

$$= 3.15 \text{ (2dp)}$$

It takes 3.15 minutes to
reach 50° .

//

(5)